INTERNATIONAL GCSE MATHEMATICS A · MODULAR · UNIT 2 HIGHER

Expected Paper 2 ELT/U2X2 Mark Scheme

Step by step, with the marks



DR ESLAM AHMED
eliteigcse.com

M = method mark, A = accuracy mark (needs the method), B = mark for a correct value on its own. Questions 19–26 also show THE IDEA: the one thought that unlocks them.

ANSWERS AT A GLANCE

1	30° angle constructed
2	18.2 minutes
3	2.5×10^5
4	36 and 60
5	$a = 3$, $b = 2$
6	2.5 litres
7	5 years
8	$n = 12$
9	region R; 9
10	$x = 20$
11	34°
12	$T = 0.2\sqrt{L}$; 1.6, 81
13	15 cm; 32 cm^2
14	13 cm
15	CF 6, 20, 45, 65, 76, 80; ≈ 43 min
16	$a = \frac{bt}{b - t}$
17	5; $\frac{x - 5}{2}$; $x = 1$, $x = -\frac{1}{2}$
18	2 seconds
19	$t = 2$, $t = 4$; 4 m
20	120 cm^2
21	9 and 4; 670
22	(2, 10); −1, 5; $k = 4$
23	**b** − **a**; 1 : 3
24	1440 cm^3
25	$x = 5$; 7.75 cm
26	$a = 3$, $b = -9$; (−3, 28)

1 Construction

2 marks

1 **60° FIRST**
Arc centre A cutting AB at X; arc of the same radius centre X meeting the first arc at Y. Angle $YAX = 60^\circ$ (equilateral triangle AXY). **M1**

2 **BISECT IT**
Equal arcs from X and Y, then join A to where they cross: 30° ($\pm 2^\circ$), all arcs shown. **A1**

2 Combined estimate of the mean

4 marks

1 **BOYS: MIDPOINTS**
$5 \times 4 + 15 \times 11 + 25 \times 9 + 35 \times 6 = 20 + 165 + 225 + 210$ **M1**

2 **BOYS TOTAL**
620 **M1**

3 **GIRLS TOTAL**
$20 \times 14.5 = 290$ **M1**

ANSWER $(620 + 290) \div 50 = 18.2$ minutes **A1**

3 Standard form

3 marks

1 **SAME UNITS**
3 cm $= 0.03$ m $= 3 \times 10^{-2}$ m **B1**

2 **DIVIDE**
$3 \times 10^{-2} \div (1.2 \times 10^{-7})$ **M1**

ANSWER 2.5×10^{5} **A1**

4 HCF and LCM

3 marks

1 **PRIME FACTORS**
$12 = 2^2 \times 3$, $180 = 2^2 \times 3^2 \times 5$, so $180 \div 12 = 15 = 3 \times 5$ **M1**

2 **SHARE THE EXTRA 3 AND 5**
each number is 12× something: 12×3 and 12×5 (12 and 180 is not allowed) **M1**

ANSWER 36 and 60 **A1**

5 Simultaneous equations

3 marks

1 **SUBSTITUTE BOTH POINTS**
$-2a + 9b = 12$ and $6a - 3b = 12$ **M1**

2 **ELIMINATE**
×3 the first: $-6a + 27b = 36$; add: $24b = 48$ **M1**

ANSWER $b = 2$, $a = 3$ **A1**

6 Ratio — which paint runs out

3 marks

1 **TEST THE BLUE**
1.2 L blue needs $1.2 \div 2 \times 3 = 1.8$ L yellow — only 1.5 L **M1**

2 **SO YELLOW RUNS OUT**
1.5 L yellow needs $1.5 \div 3 \times 2 = 1.0$ L blue **M1**

ANSWER $1.0 + 1.5 = 2.5$ litres **A1**

7 Depreciation — how many years

3 marks

1 **MULTIPLIER**
value after n years = $18\,000 \times 0.85^n$; half is \$9000 **M1**

2 **TRY VALUES**
$n = 4$: \$9396.11 $n = 5$: \$7986.70 **M1**

ANSWER 5 years **A1**

8 Polygons

3 marks

1 **EXTERIOR ANGLES**
P: $\frac{360}{n}$ Q: $\frac{360}{2n} = \frac{180}{n}$ **M1**

2 **INTERIOR DIFFERENCE = EXTERIOR DIFFERENCE**
$\frac{360}{n} - \frac{180}{n} = 15 \Rightarrow \frac{180}{n} = 15$ **M1**

ANSWER $n = 12$ (interior angles 150° and 165°) **A1**

9 Region and integer points

4 marks

1 **(A)**
The triangle to the right of $x = 1$, above $y = x$ and below $x + 2y = 10$, corners $(1, 1)$, $(1, 4.5)$, $\left(\frac{10}{3}, \frac{10}{3}\right)$, labelled R **B2**

2 (B1 for a region correct on two of the three lines)

3 **(B) INTEGER POINTS NEAR THE CORNER**
the corner $\left(\frac{10}{3}, \frac{10}{3}\right)$ is not an integer point; test $(3, 3)$: 9, $(2, 4)$: 8 **M1**

ANSWER 9 at (3, 3) **A1**

10 Up x% then down x%

3 marks

1 **MULTIPLIERS**
$\left(1 + \frac{x}{100}\right)\left(1 - \frac{x}{100}\right)$ **M1**

2 **OVERALL 4% LESS**
$1 - \frac{x^2}{10\,000} = 0.96 \Rightarrow x^2 = 400$ **M1**

ANSWER $x = 20$ **A1**

11 Circle theorems

4 marks

1 **ISOSCELES TRIANGLE**
$OA = OC$ (radii), so angle $OCA = 28^\circ$ **M1**

2 **ANGLE AT THE CENTRE**
angle $COT = 56^\circ$ — angle at the centre is twice the angle at the circumference **M1**

3 **TANGENT**
angle $OCT = 90^\circ$ — tangent is perpendicular to the radius; 180 − 90 − 56 **B1**

ANSWER 34° **A1**

4 **OTHER ROUTE**
$ACB = 90^\circ$ (angle in a semicircle), $BCT = 28^\circ$ (alternate segment), $CBT = 118^\circ$, so $ATC = 180 - 118 - 28 = 34^\circ$

12 Direct proportion

4 marks

1 **(A)**

$T = k\sqrt{L}$: $1 = k\sqrt{25} \Rightarrow k = 0.2$ **M1**

ANSWER (a) $T = 0.2\sqrt{L}$ **A1** (b) $0.2\sqrt{64} = 1.6$ **B1**; $\sqrt{L} = 9 \Rightarrow L = 81$ **B1**

13 Similar solids

4 marks

1 **VOLUME FACTOR**

same chocolate, so masses give volumes: $625 \div 40 = 15.625$

2 **LENGTH FACTOR**

$\sqrt[3]{15.625} = 2.5$ **M1**

3 **(A)**

$6 \times 2.5 = 15$ cm **A1**

4 **(B) AREA FACTOR**

$2.5^2 = 6.25$: $200 \div 6.25$ **M1**

ANSWER (a) 15 cm (b) 32 cm^2 **A1**

14 Spheres in a cylinder

3 marks

1 **VOLUME OF THE SPHERES**

$3 \times \frac{4}{3}\pi(3^3) = 108\pi$ **M1**

2 **RISE OF THE WATER**

$108\pi \div (\pi \times 6^2) = 3$ **M1**

ANSWER $10 + 3 = 13$ cm **A1**

15 Cumulative frequency

5 marks

1 **(A)**
6, 20, 45, 65, 76, 80 **B1**

2 **(B)**
points at the END of each class: (10, 6), (20, 20), (30, 45), (40, 65), (50, 76), (60, 80) **M1;**
smooth curve (or lines) through them, starting at (0, 0) **A1**

3 **(C) 12 SLOWEST**
they are above cumulative frequency 80 − 12 = 68: read across from 68 **M1**

ANSWER about 43 minutes (accept 42 to 44, or from their graph) **A1**

16 Change the subject

4 marks

1 **MULTIPLY OUT**
$t(a + b) = ab \Rightarrow at + bt = ab$ **M1**

2 ***a* TERMS TOGETHER**
$bt = ab - at$ **M1**

3 **FACTORISE**
$bt = a(b - t)$ **M1**

ANSWER $a = \dfrac{bt}{b - t}$ **A1**

17 Functions

5 marks

1 **(A)**
$g(3) = 6$, so $f(6) = 12 + k = 17$ **M1**

2 $k = 5$ **A1**

3 **(B)**
$y = 2x + 5 \Rightarrow x = \dfrac{y - 5}{2}$: $f^{-1}(x) = \dfrac{x - 5}{2}$ **B1**

4 **(C)**
$\dfrac{x - 5}{2} = x^2 - 3 \Rightarrow 2x^2 - x - 1 = 0 \Rightarrow (2x + 1)(x - 1) = 0$ **M1**

ANSWER $x = 1$ or $x = -\frac{1}{2}$ **A1**

18 Quadratic inequality in context

3 marks

1 **INEQUALITY**
$20t - 5t^2 > 15 \Rightarrow t^2 - 4t + 3 < 0$ **M1**

2 **CRITICAL VALUES**
$(t - 1)(t - 3) < 0 \Rightarrow 1 < t < 3$ **M1**

ANSWER 3 − 1 = 2 seconds **A1**

19 Kinematics — distance between two stops

5 marks

THE IDEA "at rest" means $v = \frac{ds}{dt} = 0$. Between the two stops the particle moves one way only, so the distance is the change in s.

1 **DIFFERENTIATE**
$v = 3t^2 - 18t + 24$ **M1**

2 **SET $v = 0$**
$3(t^2 - 6t + 8) = 0 \Rightarrow 3(t - 2)(t - 4) = 0$ **M1**

3 **(A)**
$t = 2$ and $t = 4$ **A1**

4 **(B) DISPLACEMENTS**
$s(2) = 8 - 36 + 48 = 20, \quad s(4) = 64 - 144 + 96 = 16$ **M1**

ANSWER 20 − 16 = 4 m **A1**

20 Rectangle — perimeter and diagonal

5 marks

THE IDEA two facts, two equations: the perimeter gives a straight line, Pythagoras on the diagonal gives a circle — solve them together.

1 **PERIMETER**
sides x and y: $x + y = 23$ **M1**

2 **DIAGONAL**
$x^2 + y^2 = 17^2 = 289$ **M1**

3 **SUBSTITUTE**
$x^2 + (23 - x)^2 = 289 \Rightarrow x^2 - 23x + 120 = 0$ **M1**

4 **SOLVE**
$(x - 8)(x - 15) = 0$: sides 8 cm and 15 cm **M1**

ANSWER $8 \times 15 = 120$ cm^2 (or $2xy = 23^2 - 289 = 240$) **A1**

21 Series from a formula for S_n

4 marks

THE IDEA S_1 is the first term, and $S_2 - S_1$ is the second term. A block of terms is a difference of two sums.

1. **(A)**
   $S_1 = 9$, $S_2 = 22$, so $u_2 = 13$ **M1**
2. first term 9, common difference 4 **A1**
3. **(B)**
   $S_{20} - S_{10} = (800 + 140) - (200 + 70) = 940 - 270$ **M1**

ANSWER 670 **A1**

22 Transformations of graphs

4 marks

THE IDEA read each transformation backwards: $f(x + a)$ through O means $f(a) = 0$, so a is a root of f.

1. **(A)**
   y values doubled: (2, 10) **B1**
2. **(B)**
   at $x = 0$: $f(0 + a) = 0$, so a is where f crosses the x-axis **M1**
3. $a = -1$ or $a = 5$ **A1**
4. **(C)**
   $kx = 2$ when $x = 0.5$, so $k = 4$ **B1**

ANSWER (2, 10); $a = -1$ or 5; $k = 4$

23 Vectors in a regular hexagon

5 marks

THE IDEA the hexagon is six equilateral triangles: $OABC$ is a rhombus, so $\vec{OC} = \vec{AB}$, and D is opposite A, so $\vec{OD} = -\mathbf{a}$.

1 **(A)**
$\vec{OC} = \vec{AB} = \mathbf{b} - \mathbf{a}$ **B1**

2 **FIND $\vec{AM}$**
$\vec{CD} = -\mathbf{a} - (\mathbf{b} - \mathbf{a}) = -\mathbf{b}$, so $\vec{OM} = \mathbf{b} - \mathbf{a} - \frac{1}{2}\mathbf{b} = \frac{1}{2}\mathbf{b} - \mathbf{a}$ and $\vec{AM} = \frac{1}{2}\mathbf{b} - 2\mathbf{a}$ **M1**

3 **X ON AM**
$\vec{OX} = \mathbf{a} + \lambda\left(\frac{1}{2}\mathbf{b} - 2\mathbf{a}\right) = (1 - 2\lambda)\mathbf{a} + \frac{\lambda}{2}\mathbf{b}$ **M1**

4 **X ON OB: NO a**
$1 - 2\lambda = 0 \Rightarrow \lambda = \frac{1}{2}$, so $\vec{OX} = \frac{1}{4}\mathbf{b}$ **M1**

ANSWER $OX : XB = 1 : 3$ **A1**

24 Surface area gives the radius

5 marks

THE IDEA the cylinder's height is $20 - r$, so the total surface area is an equation in r only — a quadratic.

1 **SURFACE AREA**
base πr^2 + curved $2\pi r(20 - r)$ + hemisphere $2\pi r^2$ **M1**

2 **SIMPLIFY**
$\pi r^2 + 40\pi r = 225\pi \Rightarrow r^2 + 40r - 225 = 0 \Rightarrow (r + 45)(r - 5) = 0$ **M1**

3 **RADIUS**
$r = 5$, cylinder height 15 **A1**

4 **VOLUME**
$\pi(5^2)(15) + \frac{2}{3}\pi(5^3) = 375\pi + \frac{250}{3}\pi = \frac{1375}{3}\pi$ **M1**

ANSWER $1439.9\ldots = 1440\ \text{cm}^3$ **A1**

25 Intersecting chords outside the circle

4 marks

THE IDEA from an outside point: $PA \times PB = PC \times PD = PT^2$. Use the WHOLE lengths PB and PD.

1 **EQUATION**
$x(x + x + 2) = 4 \times (4 + 11) \Rightarrow 2x^2 + 2x = 60$ **M1**

2 **SOLVE**
$x^2 + x - 30 = 0 \Rightarrow (x + 6)(x - 5) = 0$ **M1**

3 **(A)**
$x = 5$ **A1**

ANSWER (b) $PT^2 = 4 \times 15 = 60$, $PT = 7.75$ cm **B1**

26 Find the curve from its turning point

5 marks

THE IDEA a turning point gives TWO equations: the gradient is 0 there, AND the point is on the curve.

1 **DIFFERENTIATE**
$\frac{dy}{dx} = 3x^2 + 2ax + b$ **M1**

2 **TWO EQUATIONS**
gradient 0 at $x = 1$: $3 + 2a + b = 0$; on the curve: $1 + a + b + 1 = -4 \Rightarrow a + b = -6$ **M1**

3 **(A)**
$a = 3$, $b = -9$ **A1**

4 **(B)**
$3x^2 + 6x - 9 = 0 \Rightarrow 3(x + 3)(x - 1) = 0$, so $x = -3$ **M1**

ANSWER $y = -27 + 27 + 27 + 1 = 28$: (−3, 28) (a maximum) **A1**